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Personalized Perfume Recommendations Based on User Descriptions Using Cosine and Jaccard Similarity

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Abstract

The research proposes a perfume recommendation system using Natural Language Processing (NLP) and content-based filtering to help consumers choose perfumes in a growing market. The system analyzes descriptive user input in Bahasa Indonesia to identify preferences through keyword matching and a question-answering approach. A dataset of 24,760 perfumes with 11 features, primarily sourced from Fragrantica, was used. The method includes exploratory data analysis, preprocessing, model design based on content-based filtering with Cosine and Jaccard similarity, and evaluation using nDCG@3, Precision@3, Recall@3, Diversity, and Coverage. The system achieved strong performance in Jaccard based recommendation with a Precision@3 and Recall@3 of 0.8889 and an nDCG@3 of 0.8978, while cosine based results are also performed well. A web-based interface using Streamlit was developed to demonstrate the system's functionality. User feedback was gathered through a survey involving 31 participants, showing positive responses with average scores ranging from 4.2 to 4.6 out of 5 across usability criteria.

Peer-review under responsibility of the scientific committee of the 10th International Conference on Computer Science and Computational Intelligence (ICCSCI) 2025

Keywords: Perfume recommendation system; Content-based filtering; Natural Language Processing (NLP); Jaccard similarity; Streamlit.

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1. Introduction

1.1. Introduction

Perfume has become a significant part of Indonesian lifestyle, with the market projected to generate US$0.44 billion in revenue by 2024, with a Compound Annual Growth Rate (CAGR) of 2.59% from 2024 to 2029 [1][2][3]. The rise of local and international perfume brands with diverse scents has provided consumers with a wide variety of choices, but also contributed to decision-making challenges [4]. Factors such as personal preference, scent type, and prior experience influence purchasing decisions [5]. To address this, a recommendation system has been implemented to align recommendations with individual preferences, ensuring that consumers can choose suitable perfumes [5][6].

This study applies Natural Language Processing (NLP) and content-based filtering to build a recommendation system for perfume. The system used a keyword matching technique and a question-answering approach to identify user preferences and generate relevant recommendations [7][8]. The system-based approach is preferred over a model-based due to its accessibility, efficiency, and optimized user experience [9]. Content-based filtering is used to address the "cold start" problem where insufficient historical data hinders recommendation accuracy [10][11]. While the usage of NLP, particularly keyword detection, is integrated to analyze textual input deeply [8].

This research has aimed to create a perfume recommender system that can identify user preferences based on their responses. It used Natural Language Processing (NLP) to understand user preferences through textual description analysis [12]. Relevant keywords are extracted and used in a content-based filtering method to match user preferences with perfume attributes and generate recommendations that align [13]. It will generate three recommendations, placing the most relevant and suitable perfume at the top to aid user decision-making. Later the system's performance will be evaluated in terms of precision and relevance of the recommendations provided.

To maintain focus and maximize research output, several limitations have been set. The primary limitation has been the use of Indonesian language as the main language for data processing and user interaction, while English has been used to identify perfume notes. The dataset has been primarily sourced from Fragrantica, a reputable platform among perfume enthusiasts for reviews and fragrance information [14]. Supplementary data has been obtained from local perfume websites and e-commerce platforms such as Shopee and Tokopedia. The dataset collected has included award-winning perfumes, featuring attributes such as ID, country of origin, brand, name, usage time (day, night, or day/night), rating, perfume notes, gender, and olfactory family.

1.2. Related Works

A recommender system is a tool that assists users by providing item suggestions in the world full of available information [6].

Research conducted by Lutan and Badica (2023) has shown that question-answering-based recommender systems can enhance user experience by providing more tailored recommendations. These systems are designed to deliver personalized suggestions based on information provided by users through a series of structured questions. This study utilized 1,000 quiz inputs to simulate 1,000 users seeking suitable perfumes. Among them, 192 quiz responses did not result in any recommendations, while 782 responses yielded recommendations that considered additional user inputs. The algorithm used achieved an effectiveness rate of 78.2% when providing recommendations based on Fragrance Family, although lower effectiveness was observed in cases where only one perfume matched the user's preferences. In terms of relevance, the system reached 1.9% in scenarios with a single matching item, and 73% when using Fragrance Family as the basis for recommendation [7]. Meanwhile, a study by Zou et al. proposed a novel question-based recommendation method called Qrec, which enables users to find item suggestions interactively by answering algorithmically selected questions. Their model employs a matrix factorization algorithm to infer user perceptions and preferences, thereby learning optimal question-asking strategies. The study uses an algorithm that evaluates the informativeness of questions to determine the most effective ones [15].

The study conducted by Bag and colleagues presented two simple yet effective similarity models that consider all user rating vectors to classify relevant contexts and generate recommendations with lower computational time. These models were tested on the MovieLens dataset. The Jaccard similarity model outperformed others in terms of accuracy and effectiveness in generating recommendations that aligned with user needs, highlighting the necessity for more efficient and accurate methods in recommender systems [16]. In addition, a study by Javed and colleagues introduced two recommender system approaches: context-aware and content-based. The context-aware system filters items based on user interests, while the content-based system utilizes patterns from the World Wide Web to generate future news recommendations. The techniques employed include content-based filtering, collaborative filtering, hybrid recommender systems, web ontology languages, RDF, JAVA, machine learning, semantic mapping rules, and natural ontology languages. In this study, the content-based recommendation system suggests websites and data of interest to users based on their recent searches or preferences [10]. Lastly, a study by Mishra and Jain implemented a subject recommendation system using content-based filtering and machine learning algorithms to filter available skills and lessons online. The system provides a personalized experience for users within the rapidly evolving e-learning infrastructure [17].

2. Methodology

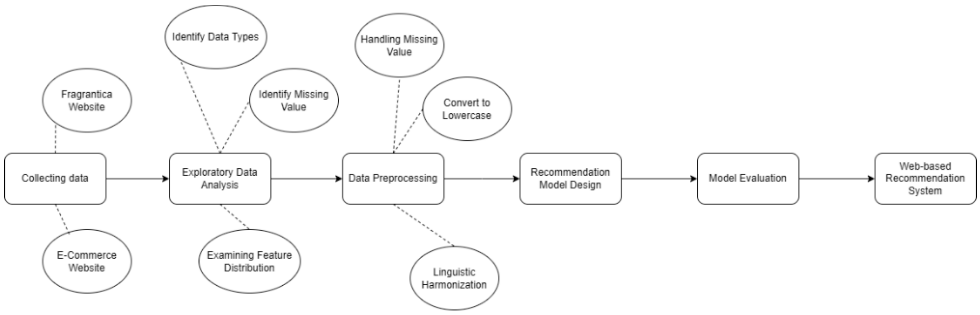


Fig. 1. Step-by-step method

2.1. Collecting Data

The dataset was collected from Fragrantica, a platform widely utilized by perfume enthusiasts for reviews and fragrance information [14]. Supplementary data had been gathered from local perfume websites and e-commerce platforms, including Shopee and Tokopedia to enhance data comprehensiveness.

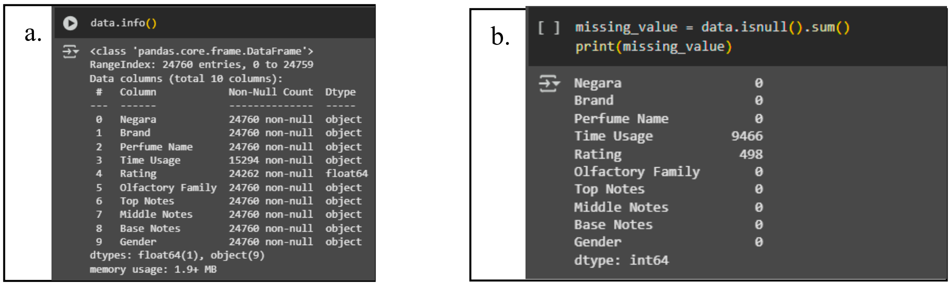
A total of 24,760 rows with 11 features had been obtained. These features include ID, country of origin, perfume brand, perfume name, usage time (day or night), rating, perfume notes, gender, and olfactory family.

Table 1. Sample Dataset

ID	Country	Perfume Brand	Perfume Name	Time Usage (Day, Night, Day/Night)	Rating	Olfactory Family	Top Notes	Middle Notes	Base Notes	Gender
128	Indonesia	Mop (mother of pearl)	savage vanilla	night	4.47	woody floral musk	rose, bergamot	cedarwood, iris	vanilla, musk	unisex
36	Indonesia	etre	volcanique	night	4.24	oriental floral	jasmine, iris	patchouli	sandalwood	unisex

2.2. Exploratory Data Analysis (EDA)

The primary objective of this stage has been to understand the characteristics, structure, and patterns within the data prior to proceeding to the subsequent phases. During this stage, issues such as missing values, data imbalance, or uneven distribution (which could potentially affect the system's performance) have been identified.



```
a.
data.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24760 entries, 0 to 24759
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Negara           24760 non-null  object
 1   Brand            24760 non-null  object
 2   Perfume Name     24760 non-null  object
 3   Time Usage       15294 non-null  object
 4   Rating           24262 non-null  float64
 5   Olfactory Family 24760 non-null  object
 6   Top Notes        24760 non-null  object
 7   Middle Notes     24760 non-null  object
 8   Base Notes       24760 non-null  object
 9   Gender           24760 non-null  object
dtypes: float64(1), object(9)
memory usage: 1.9+ MB
```

```
b.
[ ] missing_value = data.isnull().sum()
    print(missing_value)
Negara              0
Brand               0
Perfume Name        0
Time Usage       9466
Rating            498
Olfactory Family    0
Top Notes           0
Middle Notes        0
Base Notes          0
Gender              0
dtype: int64
```

Fig. 2. (a) Data Type; (b) Missing Value.

The initial step of Exploratory Data Analysis (EDA) involved identifying data types and missing values. The dataset consists of 10 string types columns and 1 float type column with 9,466 missing values found in the time usage column and 498 missing values in the rating column.

The study analyzed feature distribution to identify category imbalances in variables like rating, brand, gender, notes, and time usage, aiding in identifying data tendencies and determining appropriate preprocessing and modeling strategies.

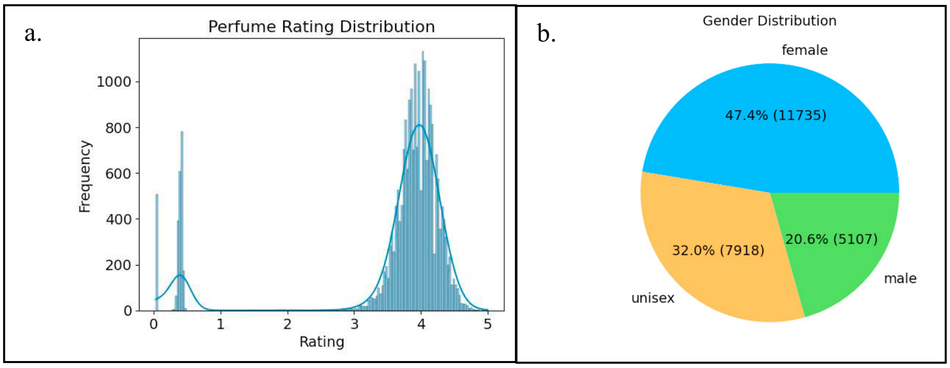


Fig. 3. (a) Perfume Rating Distribution (b) Gender Distribution

Figure 3(a) displays the distribution of perfume ratings, showing the highest frequency at a rating of 4 on a 5-point scale, with declining frequencies around ratings 3 and 5 and an unexpected spike near 0, indicating a left-skewed distribution with unusual low-rating values, while Figure 3(b) presents the gender distribution of perfumes, revealing female-categorized perfumes as the majority at 47.4% (11,734 entries), followed by unisex at 32% (7,917 entries) and male at 20.6% (5,107 entries).The last is the distribution of perfume brands.

The exploratory analysis indicates that Avon dominates the dataset in terms of frequency, with more than 600 registered perfume variants. Although Natura ranks lowest among the top 5 brands, it still makes a notable contribution, with over 200 perfume variants listed under its name.

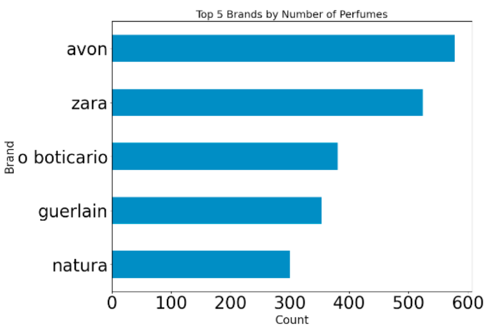


Fig. 4. Top 5 Brands

2.3. Data Preprocessing

At this stage, the data undergoes processing, organization, and cleaning to address missing values and inconsistencies. The objective of this process is to produce a structured and reliable dataset suitable for subsequent analytical procedures.

The initial step involves handling missing values through imputation. The time usage variable contains approximately 38.2% missing values. The imputation process for this variable will take into account the available notes associated with each entry.

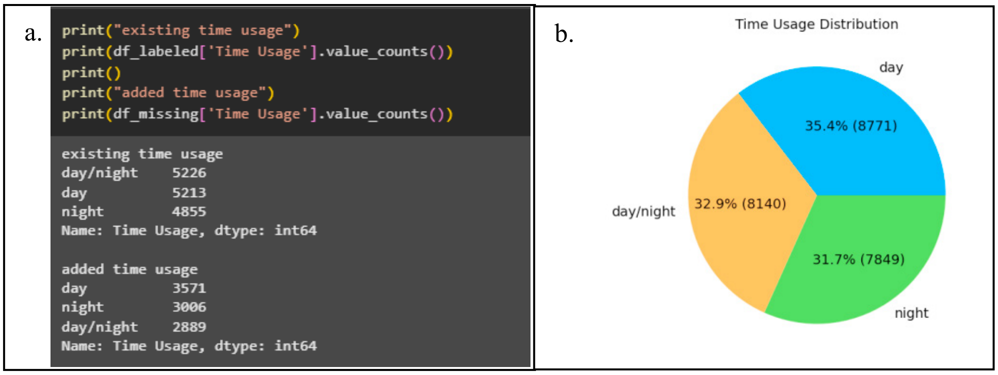


Fig. 5. (a) Existing and Added Time Usage; (b) Final Chart Time Usage Distribution.

A similar approach was applied to the rating variable, which had 2.01% missing values. Despite the low proportion, median imputation (3.93) was performed due to its significance in the recommendation system and its non-normal distribution [18][19].

The data was converted to lowercase to ensure consistency and facilitate processing during the Natural Language Processing (NLP) stage [20]. Subsequently, adjustments are made to the delimiter in the olfactory column by replacing spaces (" ") with a comma followed by a space (", "). This adjustment has aimed to bring consistency to the formatting of the aroma element, aligning it with the structure used in the notes column, where aromatic components such as pink pepper are represented as a single entity and separated by commas. This consistency in formatting has been essential for enabling seamless integration of all relevant columns into a unified feature, which will be used in subsequent analysis and recommendation system modeling.

Furthermore, linguistic harmonization was also applied by translating country names and adjusting labels in the gender and time usage columns to Indonesian. These preprocessing steps ensured uniformity and supported accurate downstream analysis.

2.4. Recommendation Model Design

This study created a Python-based perfume recommendation model that utilizes user-input features such as country, brand, name, gender, rating, notes, and olfactory family to provide accurate suggestions, incorporating keyword detection to analyze the user's desired perfume description by identifying relevant keywords and matching them against the dataset's perfume features using Jaccard and Cosine similarity metrics. Jaccard similarity is the ratio between the size of the union and intersection of the sets of samples [16]. While cosine similarity assesses the degree of similarity between sample vectors by computing the cosine of the angle formed between them [21]. Perfumes with the highest similarity score are prioritized and presented as recommendations to the user.

$$Sim(u,v)^{Jaccard} = \frac{(I_u \cap I_v)}{I_u \cup I_v} \tag{1}$$

$$SIM_{cos\theta}\left(\overrightarrow{q_t}, \overrightarrow{d_t}\right) = \frac{\overrightarrow{q_t} \cdot \overrightarrow{d_t}}{|\overrightarrow{q_t}| \times |\overrightarrow{d_t}|} \tag{2}$$

Algorithm 1 : Pseudocode for building perfume recommendation system using Jaccard and Cosine Similarity

Input: *Perfume Dataset*
Output: *Perfume Recommendation*
1. ***START***
2. LOAD dataset and preprocess (clean, vectorize features)
3. *GET user preference for similarity method (Jaccard/Cosine)*
4. GET user inputs: gender, time, description, exclusions
5. EXTRACT keywords and rating filters from description
6. EXTRACT exclusion keywords
7. FILTER dataset by:
    Gender and time usage
    Description keywords (include 'lokal' for local products)
    Rating constraints (if any)
    Excluded keywords
7. IF description keywords exist:
    IF user chose jaccard similarity:
        CALCULATE Jaccard similarity between query and items
        SORT by Jaccard similarity score
    ELSE IF user chose cosine similarity:
        CALCULATE Cosine similarity between query and items
        SORT by Cosine similarity score
8. RETURN top 3 recommendations or "no results"
9. ***END***

2.5. Model Evaluation

To evaluate the performance of the developed recommendation system, this paper has used five primary evaluation metrics such as: Normalized Discounted Cumulative Gain (nDCG), Precision@k, Recall@k, Coverage and Diversity.

nDCG (Normalized Discounted Cumulative Gain) is used to assess the quality of the recommendation ranking by considering the positions of relevant items in the list. The score for this metric increases when relevant items appear at higher ranks in the recommendation list [22].

$$DCG_p = \sum_{i=1}^{p} \frac{2^{rel_i} - 1}{log_2(i+1)} \tag{3}$$

$$nDCG_p = \frac{DCG_p}{iDCG_p} \tag{4}$$

Precision@k has measured the proportion of relevant items among the top-k recommended items. This metric has focused on how accurately the system presented perfumes that match user preferences within a limited number of suggestions [21].

$$Precision@k = \frac{\text{Number of relevant items in top-k}}{\text{k}} \tag{5}$$

Recall@k has calculated the proportion of relevant items that have appeared in the top-k recommendations compared to the total number of relevant items that should have been recommended. This metric has evaluated how well the system has captured all relevant items overall [22].

$$Recall@k = \frac{|\{\text{Relevant items}\} \cap \{\text{Top} - \text{k recommended items}\}|}{|\{\text{Relevant items}\}|} \tag{6}$$

This study also uses Catalog Coverage and Intra-List Diversity (ILD) as evaluation metrics. Coverage shows how many of the total available items are actually recommended. On the other hand, diversity is basically the opposite of similarity. It measures how different the recommended items are from one another. Higher diversity means the recommendations are more varied, which can help avoid redundancy and better match different user preferences [23].

$$\text{Catalog Coverage} = \frac{|\text{Unique Items Recommended}|}{|\text{Total Items in Catalog}|} \times 100\% \tag{7}$$

$$ILD(R) = \frac{2}{n \cdot (n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} dissimilarity(r_i, r_j) \tag{8}$$

2.6. Web-based Recommendation System

The developed recommendation model has been integrated into an interactive recommendation system. This system has been built using Streamlit to create an interactive web interface that can be easily accessed by users. Within this system, users are prompted to provide input regarding their preferences.

Two “radio” menus have been provided for user to choose whether they would use cosine or jaccard similarity on their recommender system. Another two dropdown menus also have been provided for selecting gender and time of usage, along with two text boxes that allowed users to enter a description of their desired perfume characteristics and specify any scent components they have wished to exclude from the recommendations.

Once all inputs are submitted, the system will display the top three perfumes that most closely match the user's description, presenting detailed information for each recommended perfume.

3. Results

The perfume recommendation system successfully implemented a content-based filtering approach combined with Natural Language Processing (NLP) to analyze user prefrences through textual descriptions. The system processed input queries in Bahasa Indonesia, extracted key features, and generated recommendations based on Jaccard similarity between user preferences and perfume attributes.

3.1. Model Performance

The evaluation metrics demonstrated strong performance, as summarized in Table 2.

Table 2. Evaluation results of the recommendation system.

Metric	Score -> Interpretation (Jaccard)	Score -> Interpretation (Cosine)
Avg. Precision@3	0.8889 -> ~89% of recommendations were relevant to the user's preferences	0.7778 -> ~78% of recommendations were relevant to the user's preference
Avg. Recall@3	0.8889 -> The system was able to find ~89% of all perfumes that should have been recommended	0.7778 -> ~ The system was able to find ~78% of all perfumes that should have been recommended
Avg. nDCG@3	0.8978 -> The most relevant perfume is placed in the top position	1 -> The most relevant perfume is placed in the top position
Catalog Coverage	0.0004 -> Only a small percentage of perfumes (0.04%) are recommended due to strict filtering.	0.0004 -> Only a small percentage of perfumes (0.04%) are recommended due to strict filtering
Diversity	0.8862 -> The recommended perfumes are very diverse.	0.8620 -> The recommended perfumes are very diverse

The system was also evaluated using user evaluation survey. The user evaluation survey, conducted with 31 respondents, has revealed strong validation of the perfume recommendation system's effectiveness while identifying key improvement areas. Respondents have consistently rated the system positively across all 8 golden rules metrics, with particularly high scores (4.2 – 4.6/5) for interface usability, recommendation understandability, and system comprehension [24]. However, critical analysis shows three recurring pain points: (1) users have expressed confusion about the Jaccard-Cosine methodology (mentioned in 17% of qualitative responses), (2) the interface's visual appeal has been rated as needing enhancement (noted by 19% of respondents), and (3) users have requested additional features like fragrance images (19% of responses) and longevity metrics (9%). Notably, the system has successfully achieved its core objective, with 80.6% of users agreeing that description-based recommendations outperform conventional search methods. The feedback has particularly highlighted the system's strength in generating perfume recommendations easily and quickly (mentioned in 75% of positive comments) while suggesting dataset expansion to include more brands as a crucial next step.

The system effectively handled queries, such as:

Table 3. Perfume recommendation model test cases.

Gender <input>	Time Usage <input>	Description <input>	Exclusion <input>	Perfume Recommendation Jaccard <Output>	Perfume Recommendation Cosine <Output>
Wanita (female)	Siang dan Malam (day/night)	Butuh parfum Prancis yang ada notes fruity apple (Need a France perfume with fruity apple notes)	Jangan ada wangi vanilla (No vanilla scent)	Phytoderm Pure Secret, Creed Spring Flower, Yves Rocher Moment De Bonheur Leau	Yves Rocher Moment De Bounheur Leau, Creed Spring Flower, Guerlain La Petite Robe Noire EDT
Pria (male)	Malam (night)	Parfum yang beraroma woody (woody scented perfume)	Jangan ada aroma floral dan fruity (No floral and fruity scents)	Zara Vibrant Leather Intense, Arabian Oud Blue Oud, Azzaro Wanted Tonic	Rasasi Woody, Labcitane Bois De Musc, Frank Boclet Patchouli

The model successfully processed all Bahasa Indonesia queries while maintaining high precision (0.8889) accross these test cases. Each recommendation matched the specified constraints, including gender preference, time usage, scent description, and exclusion criteria. Notably, the model demonstrated robustness in handling compound requests (e.g., Test case 2's dual exclusions) and localized terms (e.g., 'lokal' for Indonesian products).

3.2. System Implementation and User Interaction

The recommendation system has been operationalized through a Streamlit-based interface (https://perfumerecjaccard-pjzzbna6xymd3iwurcw3yh.streamlit.app/), serving as a proof-of-concept platform that enables users to interact with the content-based filtering algorithm. The system offers transparent output, ranking recommendations based on relevance, and includes a usage guide and dataset statistics. The system has incorporated key features such as input flexibility, allowing users to specify preferences through both structured inputs (e.g., gender, time of usage) and unstructured text (e.g., scent description, exclusions). It has also ensured real-time processing, delivering results in under 5 seconds by tokenizing input text, applying TF-IDF vectorization, and filtering perfumes using either Jaccard similarity or cosine similarity (users can choose their preferred method). Additionally, the system has provided transparent output by ranking recommendations based on relevance, along with clear usage guide and dataset statistics for better user understanding.

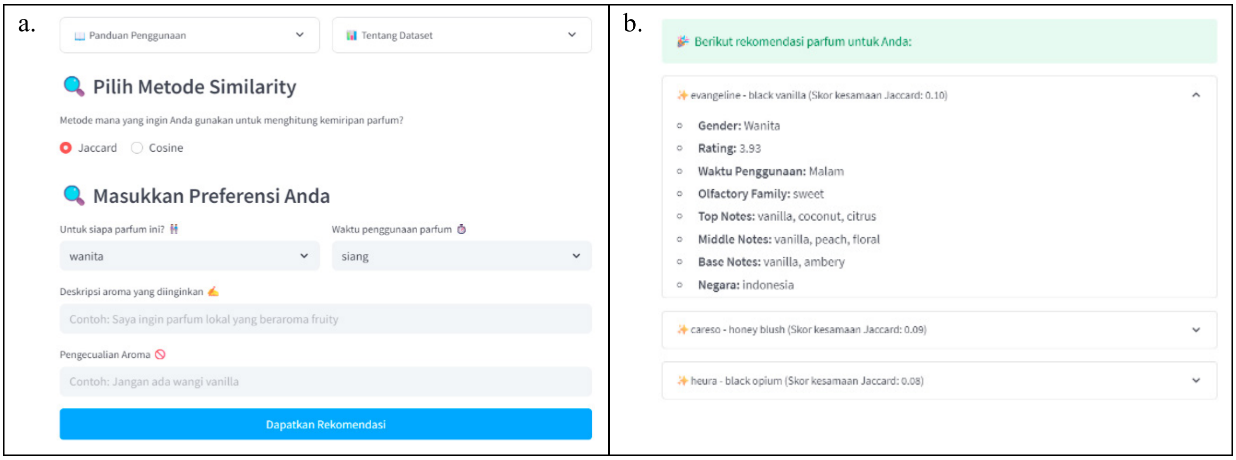


Fig. 6. (a) perfume preference input form interface; (b) perfume recommendation result.

4. Discussion

This research has shown the effectiveness of a content-based filtering approach in a perfume recommendation system. The model achieved high precision and recall scores, aligning suggestions with user preferences accurately. Natural Language Processing (NLP) was integrated for keyword extraction, enabling robust preference analysis. The system also used a question-and-answer methodology, enhancing personalized interaction and increasing user engagement. The Jaccard similarity model validated its efficacy in generating accurate recommendations while minimizing computational time. However, the system relied on Indonesian language for user input but required English for perfume notes, potentially creating accessibility barriers. Future improvements could include seasonal trends, user reviews, consideration of sillage, projection, and longevity, and multilingual support for broader accessibility. These features are expected to refine the accuracy of recommendations and accommodate a wider range of user preferences. The successful deployment of the system through a web application has demonstrated its practical applicability, creating opportunities for future advancements in personalized fragrance selection.

Data Availability

The data and source code for the development of the recommendation system can be accessed through the following link: https://github.com/rainaimtiyaz/perfume_rec_jaccard.

Author Contribution Statement

L. D. Nurmuthia and R. Imtiyaz took the lead on data collection, pre-processing, modeling, and analysis, and were also responsible for writing the manuscript. They incorporated all necessary revisions based on reviewer feedback. N. T. M. Sagala supervised the project, helped shape the research framework, and provided critical review of the manuscript to ensure its intellectual quality.

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